

HOSPITAL MANAGEMENT SYSTEM

Guide:

Raji S R (Asst.Prof.CSE)

Prepared By:

Archana C B

- MGC21CS002

Aslama Hisa

-MGC21CS005

Roshna Kabeer

-MGC21CS011

Sridhanya S

-MGC21CS014

ABSTRACT

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Our project Clinic Management System is a highly useful software for doctors as well as patients. The system aims to enhance the efficiency and effectiveness of clinic workflows, help communication between doctors and patients, and provide a seamless healthcare experience online. It consists of three modules: Admin, Doctor and Patient. It is a standalone system which can be used with valid login credentials. The admin supervises the working of doctor module, the admin can add the doctor as well as view the doctors list. The doctor can use valid login credentials which is obtained after a successful registration to login into his/her profile. Same is the process for patient as well. The doctor can monitor the medical history of patients and stay informed about their upcoming appointments. The patient can book an appointment, get information about the doctor and different departments available. By leveraging technology and streamlining administrative tasks, this system contributes to enhanced patient care, efficiency, and better health outcomes.

INTRODUCTION



ABOUT THE PROJECT

Our project Clinic Management System is a highly useful software for doctors as well as patients. The system aims to enhance the efficiency and effectiveness of clinic workflows, help communication between doctors and patients, and provide a seamless healthcare experience online. It consists of three modules: Admin, Doctor and Patient. It is a standalone system which can be used with valid login credentials. The admin supervises the working of doctor module, the admin can add the doctor as well as view the doctors list. The doctor can use valid login credentials which is obtained after a successful registration to login into his/her profile. Same is the process for patient as well. The doctor can monitor the medical history of patients and stay informed about their upcoming appointments. The patient can book an appointment, get information about the doctor and different departments available. By leveraging technology and streamlining administrative tasks, this system contributes to enhanced patient care, efficiency, and better health outcomes.

SYSTEM ANALYSIS



EXISTING SYSTEM

- Cannot view details in other devices.
 - Cannot store the details of prescription.
 - It takes too much time and efforts.
 - Availability of doctors is unknown to the patients.
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PROPOSED SYSTEM

- Doctors can access the patient medical histories from their device whenever needed.
- Highly automated and user-friendly appointment booking system.
- It offers an efficient way to search for doctors based on their specialization and book appointments.
- Provides a platform for doctors to create a profile for enabling patients to book appointments.
- Resolves the complexities of managing and booking appointments according to user preferences.

LITRACTURE SURVEY

08

AUTHOR	BOOKNAME	YEAR	PUBLICATION
Rajib Mall	Fundamentals of software engineering	2018	PHI Learning
Shari Lawrence Pfleeger	Software engineering Theory and Practice	2010	Pearson
Brad Green & Shyam Seshadri	Angular JS	2013	O'Reilly
Simon Holmes & Clive Harber	Getting MEAN with Mongo,Express,Angular, and Node	2019	Manning Publications

WEBSITE REFERENCE:

- <https://nevonprojects.com/android-patient-tracke>
- <https://devpost.com/software/patient-tracker-project>
- <https://github.com/topics/hospital-management>
- <https://tmc.kerala.gov.in/>
- <https://gg-hospital.com/>

MODULES



ADMIN MODULE

- The admin module allows admin to login and view the details of doctors and patients registered in the respective modules.
- There is a pop-up window leading to the login for admin, doctor and patient.
- The admin should click the respective button that leads to the admin login.
- The admin successfully login into his profile and he/she can view the details of registered doctors and patients.

PATIENT MODULE

- The patient module allows patients to register, login, and log out.
- The patient is one who gets the best experience from the system.
- First, patient will be visiting the home page.
- The patient should click the respective button leading to patient login.
- If not registered, the patient should click the sign-up provision on navbar and can register and login afterwards.
- The patient successfully login into his profile and he can view the details of registered doctors and can book appointment.

DOCTOR MODULE

- The doctor module allows doctors to register, login, view patient details and upcoming appointments, and logout.
- First, doctor will be visiting the home page.
- The doctor should click the respective button leading to doctor login.
- If not registered, the doctor should click the sign-up provision on navbar and register and login afterwards.
- The doctor successfully login into his profile and he/she can view the details of patients and upcoming appointments.

REQUIREMENT SPECIFICATIONS

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Hardware Requirements

- Processor : Intel core i7 Processor
 - Speed : 3GHz or above
 - RAM : 3GB or above
 - Hard Disk Capacity : 1TB
 - Keyboard : Multimedia Keyboard
 - Mouse : Standard USB : 2.0 & 3.0
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- Three white diagonal lines of varying lengths and positions are located in the bottom right corner of the slide, serving as a decorative element.

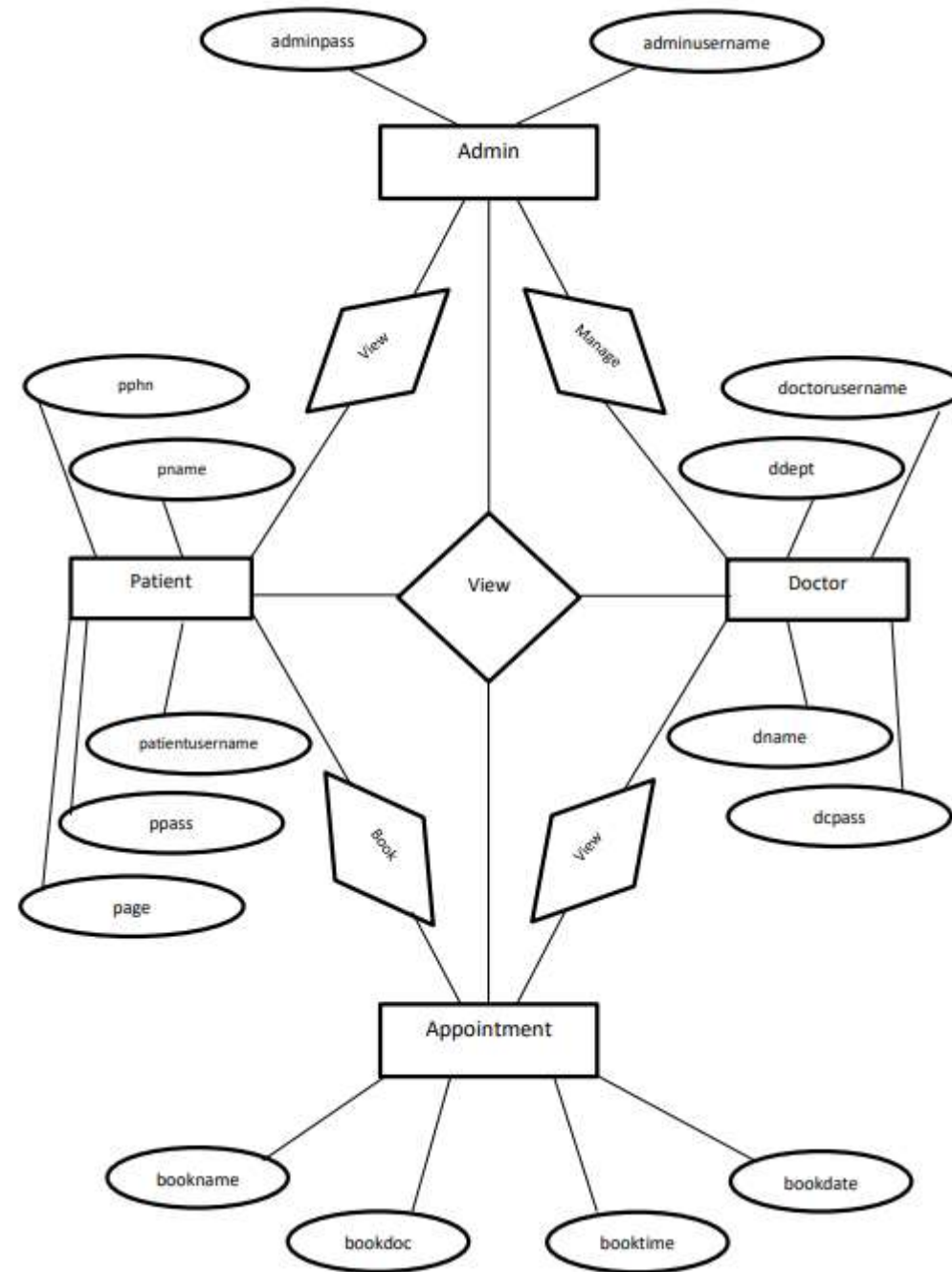
Software Requirements

- Operating System : Windows 11
 - Front End : HTML 5, CSS
 - Backend : Node JS
 - Language : JavaScript
 - Tools used : Visual Studio Code
 - Database : MongoDB
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MODELLING

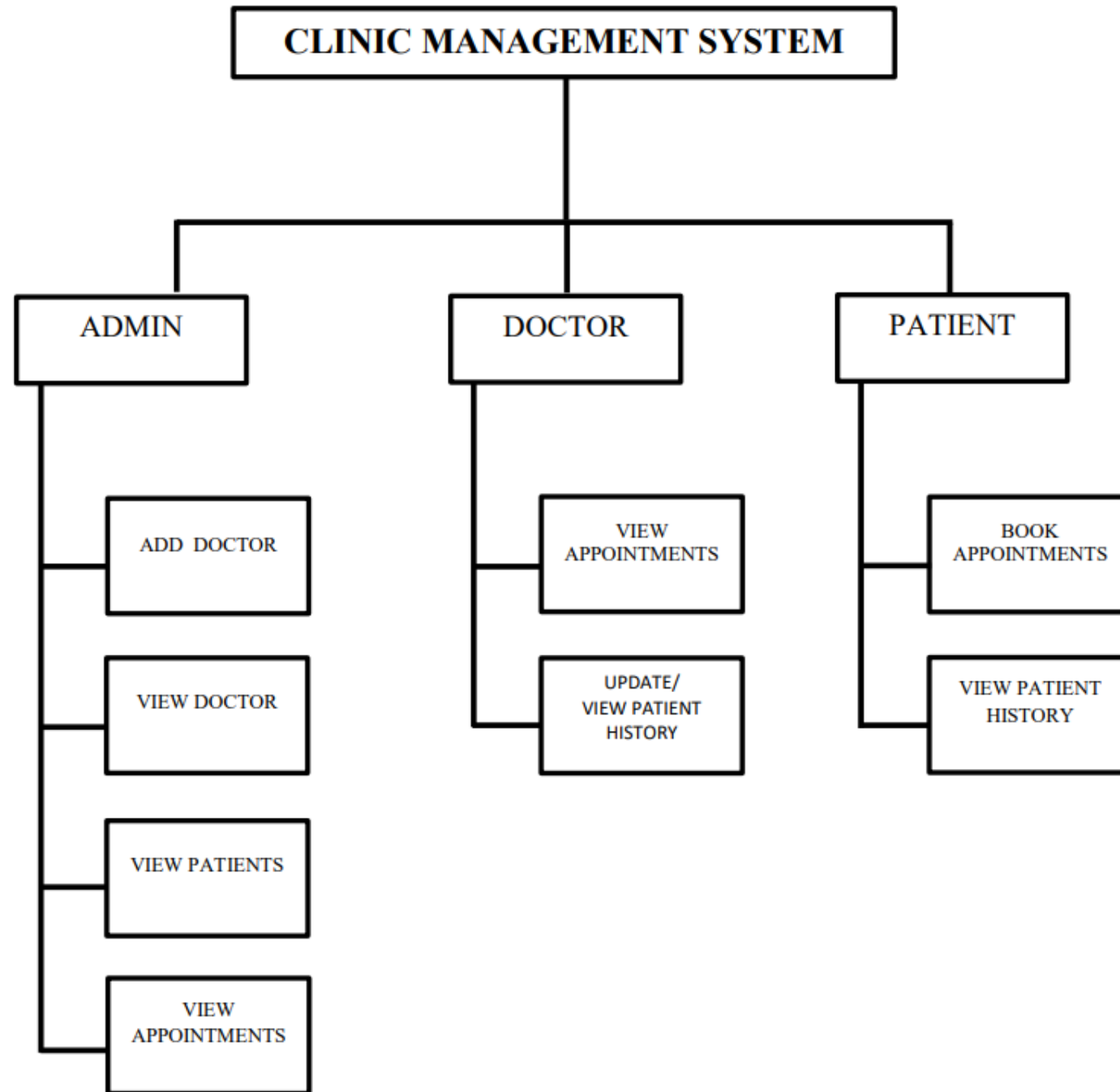


Entity relationship model



ARCHITECTURAL DESIGN

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SECURITY, BACKUP AND RECOVERY MECHANISM

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Security:

- Users must register with a unique username and a password.
- Administrators play a vital role.
- The approval process ensures only legitimate users gain access.
- It covers registration, authentication, and administrative approval.
- Administrative oversight maintains data integrity and security.

Backup:

- When users encounter issues, a robust support system steps in.
- In case of unexpected issues (data loss, bugs, system failures), the source code serves as an invaluable resource.
- Access to the source code allows for restoration and improvements, enhancing reliability and resilience.
- To ensure long-term sustainability and security, a backup source code is diligently maintained.

Recovery:

- The system prioritizes user experience by offering a simple initial response like refreshing the application, which can often resolve minor glitches swiftly.
- The system offers different ways to fix problems because not everything can be solved by refreshing.
- Users are guided through troubleshooting steps for more complex issues, ensuring they can effectively address problems on their own.
- Users have the option to report problems to customer support, allowing for personalized assistance when needed.

FUTURE ENHANCEMENT

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- The system will connect hospitals, doctors, and patients, making interactions and data sharing seamless.
- Paying bills and ordering medicine online makes healthcare easier for patients.
- Rating doctors, booking time slots, and showing availability online will make patients happier and healthcare smoother.
- Video calls let patients see doctors from home, making healthcare easier and more comfortable.

CONCLUSION



- Purpose: The software aids doctors in managing and viewing patient data conveniently.
- Installation: It's a standalone system installable on a doctor's desktop or phone for easy access.
- Data Entry: Doctors can input various patient details and medical history into the system.
- Storage and Retrieval: Patient data is saved and can be retrieved whenever necessary, searchable by name or date.
- Accessibility: The application can be used in public places with Wi-Fi and is being developed for offline usability as well.
- Availability: It will be downloadable from major app stores like Google Play and Apple Store.
- Reliability: By keeping patient and doctor records on a single platform, the system ensures reliability and efficiency.
- User-Friendly: The app is designed to be easy to use and requires minimal space while offering powerful features.
- Future Enhancements: Planned features include video conferencing, online medicine shopping, and doctor rating options to meet user needs effectively.